Rabin Dhamala

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Experience

Database Engineer *Kathmandu, Nepal*
Longshine Technology *May 2026 - Present*

- Support the Nepal Electricity Authority's large-scale data migration from the legacy Mpower platform to a new Revenue Management System (RMS).
- Analyze source and target Oracle schemas, map electricity billing, revenue collection, accounting, and financial data, and develop migration procedures.
- Write and debug SQL queries, PL/SQL stored procedures, and scripts for data transformation, validation.
- Automate exports, backups, and file transfers using bash scripting, and Python.
- Investigate migration failures, analyze logs, data inconsistencies, and troubleshoot database issues.

Software Engineer *Kathmandu, Nepal*
Uptechsys Pvt. Ltd *April 2024 - Sep 2024*

- Here, I primarily worked as a Django Backend Developer. Here, I deployed and maintained several projects. Notably, I worked on Hamro Academy, E-Nagari, Municipal Planning Management System. I also setup the blog infrastructure system. Additionally, I improved the integration and deployment workflow by integrating a testing environment and automating deployment for several of these projects.

Software Engineer *Remote, US*
Cleavr *Feb 2023 – Nov 2023*

- The work was entirely remote, requiring clear communication to ensure alignment on my plans, the team's priorities, and what was best for the company. The project involved a small, agile team of 2-3 members, where we handled everything from frontend and backend development to writing blogs and providing customer support. The domain required a solid understanding of DevOps practices, and over time, I gained valuable knowledge in areas such as **SSL**, **Nginx**, **DNS**, **monitoring tools**, and **system services**. Bash scripting was essential for setting up these services effectively, and the experience significantly deepened my technical expertise.
- The tech stack involved **AdonisJs** on the backend and **Vue** on the frontend following a monolithic architecture connected via **InertiaJs**. I worked on a variety of issues, such as providing support for new versions of database such as Postgres, Mariadb, fixing UI respnsiveness issues in frontend, integrating a complete **email change system**, enabling **server restart** via web portal, and adding feature such as **disk cleanup** etc.

Software Engineer *Kathmandu, Nepal*
Aayulogic Pvt.Ltd *Oct 2021 – March 2023*

- Here, I specialized in **Python**, working mainly in **Django** and **DRF** to build **APIs** for the frontend. I worked on attendance penalty system, optimizing payroll, transforming static recruitment process to dynamic. Technically, I gained knowledge in implementing and maintaining authentication system, **RBAC** system, learnt and used **Docker** in our project workflow, implemented caching, used multiprocessing to parallelize CPU intensive tasks, used threading (**Django Q**) for asynchronous tasks like email and notification.

Education

Pulchowk Campus, Tribhuvan University, Nepal *Nov 2016 – Sept 2021*
Bachelors in Electrical Engineering

- scored 72.5 % overall
- Notable courseworks on C, C++, Numerical Methods, Microprocess 8085, 8086, System Reliability and Artificial Neural Network

Technical Growth

- **Data structure and Algorithm**: After taking the break from work, I primarily focused on DSA, solving over 900 LeetCode problems and reaching a 1770 contest rating. This is a self repository, DSA. The objective was to improve problem solving ability by working on DP problems, binary search algorithm, graph problems, and searching algorithms.
- **Codecrafters challenge**: I completed several codecrafters challenge, including building http server, regex engine, Redis, DNS server, Docker program to pull image from Docker hub, building your own Git on early 2024. The main aim was to learn the underlying technology by implementing some part of the tool itself. I have spent the the last 3-4 months (Oct 2025 to Feb 2026) again working on Redis, Interpreter, creating your own Shell,. The aim was to be fairly comfortable with low level programming in "Go".
- **Deep Learning Specialization course by Andrew Ng on Coursera**. I learnt about neural networks (CNN, RNN) and implemented them using Tensorflow. Deep learning Specialization
- **Career break (Oct 2024 - Apr 2026)** Took a planned year-long break to prepare for master's studies. During this period, I deepened my understanding of operating systems, distributed systems, computer networks, and databases through self-study. Although I wasn't able to pursue studies abroad as planned, the experience strengthened my technical foundation and reinforced my commitment to returning to professional software development with renewed focus and drive. However, it would be in my bad faith to paint a rosy picture of this but I believe I am a better SWE having gone through this period.

Programming Skills

- **Languages**: Python, JavaScript, Node.js, Go, SQL, PL/SQL, C/C++, Bash
- **Frameworks and Technologies**: Oracle Database, Linux, Docker, Django, Vue, Redis, Nginx, PostgreSQL